The density of Garcia's quadratic-residue jumps

Principia Math In Progress

29 September 2026

Task 1 (original problem). For a positive integer p, let $r_p(t)$ denote the least nonnegative residue of the integer t modulo p. Garcia [1, pp. 5, 22] defines, for $p = 4n - 1$,

$$\Gamma_n(k) = r_p((k-1)^2) + r_p(k + 1 - 3n), \qquad J_n = \#\{k \in \mathbb{Z} : 2 \le k \le n + 2,\ \Gamma_n(k) \ge p\}.$$

MathDB 357172, recording [1, Conjecture 6.5, p. 29], asks whether $J_n/n \to 3/8$ as $n \to \infty$ through integers for which $4n - 1$ is prime.

Theorem 1. *With these definitions, $J_n/n \to 3/8$ as $n \to \infty$ through* all *integers. In particular the prime-subsequence conjecture holds.*

We give all approximation details needed for the proof. Write $\mathrm{e}(t) = \exp(2\pi i t)$.

Lemma 1. *For integers $q \to \infty$, the points*

$$z_{q,j} = \left(\frac{j}{q}, \frac{r_q(j^2)}{q} \right), \qquad 0 \le j < q,$$

are uniformly distributed in the unit square: the proportion in any fixed polygon tends to its area. Including or excluding boundary segments does not affect this limit.

Proof. For fixed integers a, b, set

$$S_q(a,b) = \sum_{j=0}^{q-1} \mathrm{e}\left(\frac{bj^2 + aj}{q} \right).$$

If $b = 0$ and $a \neq 0$, the geometric sum is zero once $q > |a|$. If $b \neq 0$, putting $u = j - k$ modulo q gives

$$\begin{aligned} |S_q(a,b)|^2 &= \sum_{u \bmod q} \mathrm{e}\left(\frac{bu^2 + au}{q} \right) \sum_{k \bmod q} \mathrm{e}\left(\frac{2buk}{q} \right) \\ &= q \sum_{\substack{u \bmod q \\ q \mid 2bu}} \mathrm{e}\left(\frac{bu^2 + au}{q} \right). \end{aligned}$$

There are $\gcd(2b, q)$ terms in the last sum, so

$$\frac{|S_q(a,b)|}{q} \le \sqrt{\frac{\gcd(2b,q)}{q}} \le \sqrt{\frac{2|b|}{q}} \longrightarrow 0. \tag{1}$$

Here the gcd uses the absolute value of its first argument. Thus the average over these points of every nonconstant Fourier monomial $\mathrm{e}(ax+by)$ tends to zero, and the constant average is one.

We justify passage from monomials to polygons. On the circle $\mathbb{R}/\mathbb{Z}$, the Fejér kernel

$$K_H(t) = \frac{1}{H}\left|\sum_{\ell=0}^{H-1} \mathrm{e}(\ell t)\right|^2$$

is a nonnegative trigonometric polynomial of integral one. For circle distance $\|t\| \geq \delta > 0$, with $\delta < 1/2$,

$$K_H(t) \leq \frac{1}{H \sin^2(\pi\delta)}.$$

This follows by summing the geometric progression. If f is continuous and periodic on the square, its convolution with $K_H(x)K_H(y)$ converges uniformly to f. Indeed uniform continuity controls the integral on $\|x\|, \|y\| < \delta$, and the displayed bound makes the remaining integral tend to zero; the product kernel has integral one. The convolutions are trigonometric polynomials. The Fourier conclusion therefore implies

$$\frac{1}{q}\sum_{j=0}^{q-1} f(z_{q,j}) \longrightarrow \int_{[0,1]^2} f(x,y)\,dx\,dy. \tag{2}$$

A polygon, viewed on the torus, has boundary contained in finitely many line segments, including any segments on the sides of the square. The area of an ε-neighborhood of this boundary tends to zero. Continuous lower and upper approximations to its indicator can be formed by linear ramps in the distance to the polygon and its complement. They agree with the indicator outside that neighborhood, take values in $[0,1]$, and have integrals converging to the area. For a closed polygon E, explicit choices are

$$f_\varepsilon^-(z) = \min\{1, d(z,E^c)/\varepsilon\}, \qquad f_\varepsilon^+(z) = \max\{0, 1 - d(z,E)/\varepsilon\},$$

where distance is on the torus. They satisfy $f_\varepsilon^- \leq \mathbf{1}_E \leq f_\varepsilon^+$. Apply (2), then let $\varepsilon \downarrow 0$. The same boundary-neighborhood bound handles any choice of boundary inclusion. This proves the lemma. □

Proof of the theorem. Take $n \geq 2$ and set $j = k - 1$. For $1 \leq j \leq n+1$,

$$-p < j + 2 - 3n < 0, \qquad r_p(j+2-3n) = j + n + 1.$$

Consequently the exact defining inequality for a jump is

$$r_p(j^2) + j \geq 3n - 2, \qquad 1 \leq j \leq n+1. \tag{3}$$

In the unit square define

$$a_p = \frac{n+1}{p} = \frac{1}{4} + \frac{5}{4p}, \qquad b_p = \frac{3n-2}{p} = \frac{3}{4} - \frac{5}{4p},$$

and

$$E_p = \{(x,y) \in [0,1]^2 : 0 \leq x \leq a_p,\ x + y \geq b_p\}.$$

The point $z_{p,0} = (0,0)$ is outside E_p, and (3) proves the exact identity

$$J_n = \#\{0 \leq j < p : z_{p,j} \in E_p\}.$$

This expression includes all endpoints with the same inequality as the original definition.

To handle the moving region, fix $0 < \varepsilon < 1/8$ and put

$$E = \{(x,y) \in [0,1]^2 : 0 \le x \le 1/4,\ x+y \ge 3/4\},$$
$$E^\varepsilon = \{(x,y) \in [0,1]^2 : 0 \le x \le 1/4+\varepsilon,\ x+y \ge 3/4-\varepsilon\}.$$

For $5/(4p) \le \varepsilon$ we have $E \subseteq E_p \subseteq E^\varepsilon$. Both bounding regions are fixed polygons, so Lemma 1 gives

$$\operatorname{area}(E) \le \liminf_{n\to\infty} \frac{J_n}{p} \le \limsup_{n\to\infty} \frac{J_n}{p} \le \operatorname{area}(E^\varepsilon).$$

Their areas are

$$\operatorname{area}(E) = \int_0^{1/4} \left(\frac{1}{4} + x\right) dx = \frac{3}{32}, \qquad \operatorname{area}(E^\varepsilon) = \frac{3}{2}(1/4+\varepsilon)^2.$$

Letting $\varepsilon \downarrow 0$ proves $J_n/p \to 3/32$. Since $p/n = 4 - 1/n \to 4$, it follows that $J_n/n \to 3/8$. □

Scope and verification. The theorem uses the original definition of jumps and proves the full conjecture. No unproved source identity, primality, or assumption of independence between the two coordinates is used. The joint distribution is proved by the mixed Fourier sums above. Exact finite checks compare the original count with the transformed inequality and polygon count; they do not replace the analytic convergence argument.

Task 2 (proposed follow-up). Find an explicit power-saving error bound for $J_n - 3n/8$, uniformly over all integers n.

Task 3 (proposed follow-up). Determine how prime powers and square divisors of $4n-1$ affect the largest fluctuations of $J_n - 3n/8$.

Task 4 (proposed follow-up). Determine limiting jump densities when the interval endpoint and the linear residue term are replaced by general fixed proportions of p.

Task 5 (proposed follow-up). Investigate corresponding joint-distribution and jump-count results for higher-degree polynomial residues, including which composite modulus sequences permit such limits.

References

[1] J. Garcia, *Class Number of the Imaginary Quadratic Field and Quadratic Residues Identities*, arXiv:2301.02951v1 (2023), 31 pages, `https://arxiv.org/abs/2301.02951`.